

EFFECTS OF GENDER, TOPOLOGY, AND TASK COMPLEXITY ON THE USABILITY OF DICTIONARY APPS FOR AN ELDERLY COHORT

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Abstract: This study investigated user task performance according to gender, interface design (linear or mixed type), and task complexity (easy, medium, or difficult). In total, 48 participants (24 females, 24 males; mean age = 64.4 years, SD = 3.04) were recruited. Task performance was indexed by the time taken to complete a task. It was noted that the main effects of gender and mode on task performance were not significant, and neither were the gender interaction effects with all other factors. It was also found that the main effects of topological structure and task effect, and their interaction, were significant with respect to task performance. An example of a good interface design (mixed type) that can be adapted to suit particular task conditions and allows high-quality operation by elderly users was proposed by this study. The mixed topological design improved task performance relative to a linear topological interface design, regardless of task complexity. The findings of this study may contribute to improvements in digital dictionary app interface design and research for elderly people.

Keywords: *Gender, Topology, Task Complexity, User-Interface Design, Digital Dictionary Apps.*

1. Introduction

English language learning is a popular activity among elderly people in Taiwan [1]. Unlike traditional learning, “mobile learning,” based on smart phone apps, enables more convenient and flexible learning [2]. A digital dictionary – one example of this technology – has been installed on most smart phones used by Taiwanese individuals. The digital English dictionary includes an input interface and audio, and has been extremely effective. Although various layouts and topological structures have been developed, it is important that tests of the interactive effectiveness and quality of the interface include elderly individuals [3]. A previous report suggested that declines in memory, attention, executive functions, and processing speed in elderly people lead to difficulties in the performance of basic daily activities [4]. An effective app interface design may help elderly individuals to more readily use products, and could result in improved quality of life and a longer duration of independence. Thus, addressing app interface

design issues based on the evaluation of elderly operational test performance is important. This study assesses the issues of gender, topology, and task complexity with respect to the usability of dictionary apps by elderly subjects; it will benefit related research and the design and evaluation of interfaces for elderly people.

1.1. Interface Design in Dictionary Apps

An investigation of digital dictionaries currently available on the Taiwanese market revealed three main types of digital dictionary: a digital translator, an online translator, and a dictionary app. Before the smart phone was popularized, digital translators were the most convenient English learning-assistance devices. Since the rapid development of the smart phone, online translator and dictionary apps are now more popular than digital translators. Indeed, dictionary apps are more accessible than other digital dictionaries, because most are free of charge and do not require online access. Table 1 presents a summary of nine dictionary apps on the Taiwanese market.

Table1. Dictionary Apps for Smart Phone in Taiwan

Name	Char.	Langu.	Input	Out put
60 lang.& pronun.	Translation can be sent by text mail.	60	Voice & Touch pad	Audio & Text
I Handy Translator Free	sent by text mail, FB., or SMS..	52	Voice & Touch pad	Audio & Text
Vocre Free Translator	Automatic oral translation.	23	Voice	Audio
Google Translator	Text reader and translation	65	Image, Voice & Touch pad	Audio & Text
Worldictionary	Text reader and translation	60	Image	Audio & Text
ColorDict	various resources for searching.	online	Touch pad	Audio & Text
Dr. eye mobile2.1	various resources for searching	online	Touch pad	Audio & Text
Dr. eye pro	various resources for searching	online	Touch pad	Audio & Text
Dr. eye 1.0	various resources for searching	online	Touch pad	Audio & Text

(summarized by the authors)

Three input methods are used by these apps: key-in via touch pad keyboard, voice, and optical character recognition (OCR) using the smart phone's camera. It was found from interviews (Fig. 1) and a related test report [5] that a touch pad keyboard was more familiar and easier to use for elderly, native Chinese subjects compared to the other two input methods. As an investigation of existing dictionary apps, three interface designers and two researchers, with more than 15 years' experience in their professional domains, were invited to an expert meeting, at which three main functional areas of the dictionary app interfaces were identified (Table 2): display, status, and input. The display function shows the output information and has three sub-functions: acting as a dictionary, facilitating learning, and providing other service functions (e.g., searching records, news, and the Internet). Finally, the input area provides a keyboard or a handwriting input function.

Most apps possess an automatically rotating display function for the smart phone that affects usability and user experience. The smart phone senses and rotates the display in the vertical or horizontal orientation according to the user's movement. Consequently, a dictionary app must switch between vertical and horizontal displays. It was noted that some elderly people preferred the vertical display because it was easier to hold, whereas others preferred the horizontal display due to its wider input area.



(a) interview of conceptual paper models



(b) evaluation of the interface proposals of topologies

Figure 1. Interview and evaluation of interface proposals of topologies**Table2. The three main functional areas of the dictionary interface**

(1) Display	(2) Status	(3) Input

1.2 Individual Differences

Xie [6] reviewed related studies and reported a main effect between young and elderly people with regard to the ability to learn how to use digital devices, based on previous learning experiences. First, people born in an era in which computer technology was already widespread may have less difficulty than elderly people, who lack a background in computer use, in operating these devices [7]. Second, increased age itself may lead to difficulties with respect to interface operation, unless appropriate feedback or support is provided. In addition to individual differences in interface operation performance, a possible significant influence of gender has also noted, particularly with regard to wayfinding capacity; research suggests a cognitive difference between men and women that could result in performance differences when performing interface operation tasks [8]. However, the effects of such differences on elderly users remain unknown. Nevertheless, elderly people are likely to be more willing to use a smart phone app if software designers design more appropriate interfaces [9].

1.3 Topology and Task Complexity

Many interface designs have been proposed. They can be categorized into three main types: linear, hierarchical, and network structures (Fig. 2) [10]. A linear topology consists of linear links: a rigid linear linking function structure clearly guides the user, but its tedious forward- and backtracking operations are more

time-consuming (Fig. 2-a). Similarly, a hierarchical topology with a layered function categorizes linear functions according to a parent-child structure (Fig. 2-b). A hierarchical structure involves a smaller memory load than a linear topological structure, but the user may become disorientated because of the divergent linear branches. In contrast, a network structure provides all functions and layouts within the same display (Fig. 2-c). Users can see a macro-view of the functional structure, but too much information on a single display may result in cognitive overload, with users becoming disorientated when attempting to locate a specific function or button.

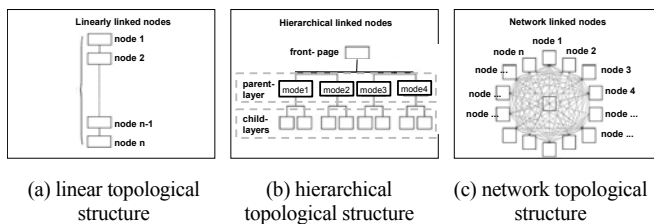


Figure2. Three main typologies: linear, hierarchical, and network structures

A previous study argued that task complexity was a major factor affecting comparisons and evaluations of interface topological structures [11]. As task complexity increases, the number of linking functions increased, along with the response time needed to locate the target function link. A multi-node task type, which requires cross-referencing information from more than one link node, is similar to a network topological structure, thus, the searching time and number of opened pages both decrease. In contrast, a single-node task requiring information from only a single link node is similar to a linear topological structure, and faster navigation performance can be predicted. A previous study showed that user performance did not differ between hierarchical and network topologies, suggesting the superiority of a “mixed topology” incorporating the best features of both [10]. However, it is important to note that repeated testing and modification of the interface is necessary to ensure the best-quality mixed design. As only one previous, small study proposed and compared linear and mixed topologies, much research remains to be conducted. In the present study, typical linear (Fig. 3) and mixed (Fig. 4) topological interface designs were proposed after a repeated design, pilot test, and redesign process.

The hypotheses were:

H1: Gender will significantly affect the comparative

ratings of digital dictionary interface usability given by elderly people: men will take less time than women to complete interface tasks.

H2: Topology will significantly affect the comparative ratings of digital dictionary interface usability given by elderly subjects.

H3: Gender, task type, and topological structure will have an interaction effect on interface operation performance.

2. Method

2.1 Participants

In total, 48 participants (24 females, 24 males), who passed a standard visual test and had previous smart phone experience, were invited to take the dictionary apps test. All participants reported 16/20 corrected visual acuity or better, and their color vision was tested using the Ishihara color plates test. No participant showed any color deficiency. All of the participants were retirees, recruited from the Activity Center of the Elderly, North Branch of Taoyuan County, Taiwan, who used a smart phone every day. Their ages ranged from 60 to 70 years, with a mean age of 64.4 (SD = 3.04) years. Participants were assigned randomly to one of the between-subject factor (gender and topological structure) experimental conditions. All participants were paid \$10 (USD) per hour for their participation.

2.2 Materials

The four independent variables were gender, mode, topological structure, and task complexity. The distributions of each variable were as follows:

- Gender: all 48 participants were distributed randomly to achieve an equal split in the sample (24 females, 24 males).
- Topological structure: the digital dictionary user interfaces were designed based on an expert meetings and a functional investigation of the study requirements for dictionary apps. Three major user-interface functions were defined according to the results of a preliminary investigation: dictionary, learning, and other functions. The number of functional nodes was 24. Each type of topology (linear and mixed) was allotted 24 subjects (12 females, 12 males) using an intuitive random principle; topology was treated as a between-subjects variable. Four nodes were used for the status area for the linear-type interface structure, and only one function could be seen and operated at

any given time (Fig. 3-a). Users could shift to other functions by clicking on the linear-linked back or forward buttons (status area of Fig. 3-b). Nine connective nodes were shown for the mixed-type interface structure, which used a mixture of network and hierarchical layouts (Fig. 4-a). Although the same functions included in the linear type were provided, the three function group layers (dictionary, learning, and other services) were displayed hierarchically, at the same time, and the nine sub-layered functions were network-structured. Users could choose to operate a specific function group by clicking on the related hierarchical function links and buttons, and the sub-groups of the six network functions were activated for further movement (status area of the Fig. 4-b).

c. Mode: every interface design proposal must possess vertical (Fig. 5-a) and horizontal display modes (Fig. 5-b). The design principle was that most of the displayed functions should be of the same size, and only the input area size was enlarged horizontally. Display mode was a within-subject variable for all tasks, so the tasks were performed twice (once in vertical, and once in horizontal, mode) with a Latin-square random order to avoid order and learning effects.

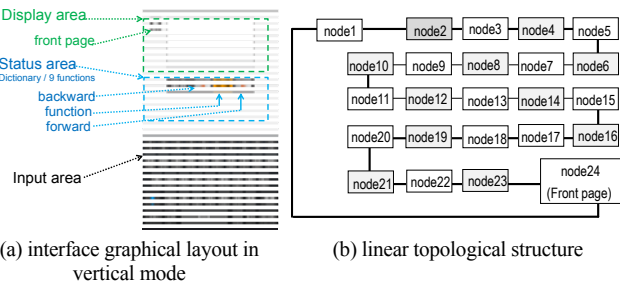


Figure 3. Linear type of topological structured interface design for the dictionary app

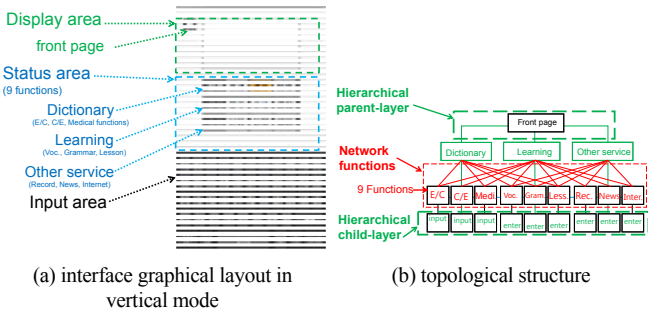


Figure 4. Mixed type of topological structured interface design for the dictionary app

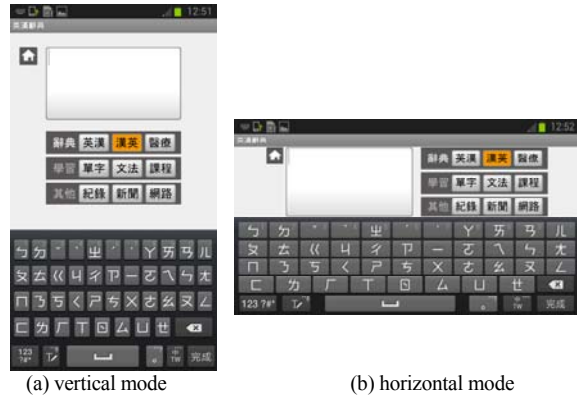


Figure 5. Vertical and horizontal display modes as an example of a mixed topological structure

- d. Task complexity: Two tasks, for each of the three task types (easy, medium, and difficult), were developed, thus yielding six tasks in total (Table 3). Subjects were required to perform the requested functions, and to adjust the relative level of the specific function, to complete the task. The specific task types were as follows:
- Easy: travel through a single line of four linking nodes and find a specific page that provided answers to the given questions.
 - Medium: travel through two targeted layers of eight linking nodes and find two specific pages that provided answers to the given questions.
 - Difficult: travel through three targeted layers of 12 linking nodes and find three specific pages that provided answers to the given questions.

Table 3. Summary of task specifications

No.	Name/ level	layers	Nodes
1	easy	Single-layer	4
2			
3	medium	Two-layer	8
4			
5	difficult	Three-layer	12
6			

2.3 Experimental design

A $2 \times 2 \times 2 \times 3$ mixed-factor experimental design was used. The independent variables were gender, topological structure (linear and mixed), mode (vertical and horizontal), and task type (easy, medium, and difficult). The dependent variable was mean time to complete the tasks. A between-subjects design was adopted for gender and topology, and a within-subject design was used for the mode and task type variables.

A multimedia room was used as the experiment site because it was quiet. The Samsung GALAXY Note II smart phone was used for the dictionary apps test. The

screen size of the projection area was 5.5" with a resolution of $1,080 \times 720$ pixels (HD).

2.4 Procedure

Participants were assigned randomly to interact with the topological structures. The nature of the experiment and the task specifications were explained briefly to each participant; they were given 15 min to learn how to interact with the user interface. More time was given to participants who felt they needed further practice. The learning and practice sessions ended when participants had completed six randomly assigned practice tasks. Instructions for these tasks were given in writing on separate cards; participants took one card at a time and completed the assigned task before continuing on to the task on the next card. This process continued until all tasks were completed. Once the practice session ended, the formal experiment began after a 15-min rest period. The tasks in the formal test were also given to participants on separate cards, and participants completed the task on one card before looking at the next. All tasks were performed randomly to avoid any carry-over effect. For both the practice and experimental trials, participants were instructed as follows: "Click on/off with your finger to activate the interface: the test will then formally begin and be recorded." Operation time indexed the effectiveness of the interface assigned, so the participants paid careful attention while performing the requested procedures. After reading and memorizing the tasks, they were asked to complete them as quickly as possible. The full experiment lasted 1.5–2 h for each participant. Participants were asked to sit in front of the smart phone at a distance of 30–40 cm (Fig. 6), and their activity was monitored throughout the experiment.

2.5 Data analysis

Task performance was indexed by the time taken to complete the task. The task ended when the participants arrived at the correct answer. The accuracy of each answer was confirmed by the research staff before the next task was started. Post hoc comparisons were performed after multivariate analysis of variance (MANOVA) to identify simple effects among factors. Statistical analyses were performed using SPSS software (ver. 20; SPSS Inc., Chicago, IL, USA.). The null hypothesis was rejected if the p-value was ≤ 0.05 .

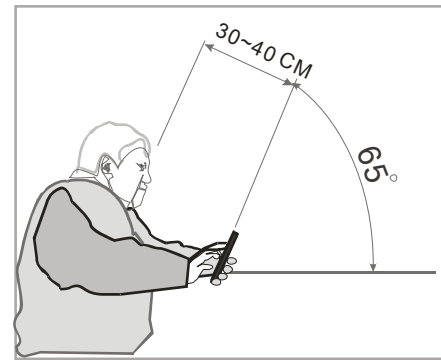


Figure6. The figure of situated participants and display

3. Results

3.1 Overall results

As Table 4 displays, the summary of the descriptive statistics of the experimental data, Table 5 further presents a summary of the MANOVA analysis of all experimental data.

Table 4. Summary of Descriptive Statistics

Variable	Levels	No.	mean	(Std. err)
Gender	female	24	37.39	(3.43)
	male	24	41.13	(3.99)
interface	linear	24	47.33	(2.49)
	mixed	24	31.19	(2.49)
mode	horizontal	48	40.48	(1.98)
	vertical	48	37.63	(2.03)
Task complexity	easy	48	16.75	(1.12)
	medium	48	41.81	(2.02)
	difficult	48	59.22	(2.83)

Table5. Summary of MANOVA analysis

		Type III Sum of Squares	df.	Mean Square	F	Sig.
Between-Sub.	Sex (A)	703.93	1	703.93	.79	.37
	Topology (B)	20124.74	1	20124.74	22.65	.00
	A × B	19.37	1	19.37	.02	.88
	S(AB)	40868.95	46	888.45		
	Mode (C)	727.58	1	727.58	2.28	.13
	A × C	251.99	1	251.99	.79	.37
	B × C	282.70	1	282.70	.88	.35
	A × B × C	154.08	1	154.08	.48	.49
	S(C)	14627.92	46	317.99		
Within-Sub.	Task type (D)	89712.86	2	89712.86	331.07	.00
	A × D	402.38	2	402.38	1.48	.22
	B × D	5395.65	2	5395.65	19.91	.00
	A × B × D	6.75	2	6.75	.02	.87
	S(D)	12464.79	92	270.97		
	C × D	9.82	2	9.82	.05	.81
	A × C × D	235.41	2	235.41	1.33	.25
	B × C × D	176.33	2	176.33	.99	.32
	A × B × C × D	2.87	2	2.87	.01	.89
	S(CD)	8144.04	92	177.04		

Topological structure ($F_{(1,46)} = 22.65$, $p < 0.001$) and task type ($F_{(3,92)} = 331.07$, $p < 0.001$) showed significant

effects, whereas gender ($F_{(1, 46)} = 0.97, p > 0.05$) and mode did not ($F_{(1, 92)} = 2.28, p > 0.05$). A post hoc comparison of the main effect analysis indicated that the mixed structure was superior to the linear topological structure. The order of time spent on the three types of task was easy < medium < difficult, indicating that the more difficult tasks took more time.

3.2 Post hoc analysis of interactions

A significant two-way interaction was detected between topology and task type ($F_{(2, 92)} = 19.91, p < 0.001$). However, the two-way interactions between gender and topology ($F_{(1, 46)} = 0.02, p > 0.05$), gender and mode ($F_{(1, 46)} = 0.79, p > 0.05$), gender and task type ($F_{(2, 92)} = 1.48, p > 0.05$), topology and mode ($F_{(1, 46)} = 0.88, p > 0.05$), and task type and mode ($F_{(2, 92)} = 0.05, p > 0.05$) were not statistically significant in the three- or four-way interactions among factors (Table 4).

Figure 7 shows the mean task completion times for the two topological structures according to the task type condition. The mean times to complete the three types of operation task were as follows: linear ($M = 20.1, SD = 9.78$) and mixed ($M = 12.9, SD = 5.00$) topological structures at the easy task level; linear ($M = 48.5, SD = 15.56$) and mixed ($M = 34.3, SD = 12.22$) topological structures at the medium task level; and linear ($M = 72.8, SD = 22.00$) and mixed ($M = 44.9, SD = 16.91$) topological structures at the difficult task level.

Results of post hoc comparisons using Tukey's HSD test indicated that significantly less time was taken with the mixed topological interface design versus the linear topological interface. The differences in time taken between the linear and mixed type interfaces were as follows: 7.2 s ($p < 0.01$) for the easy task, 14.2 s ($p < 0.01$) for the medium task, and 27.9 s ($p < 0.001$) for the difficult task. A significant positive interaction was detected between interface design and task complexity. The level of significance of the effect of topology increased with task complexity.

4. Discussion

4.1. Main effects

The experimental data analysis showed that the main effects of gender and mode on task performance were not significant, and neither were the gender interaction effects with all other factors. This is in contrast to several previous studies that reported a significant gender effect [12, 13]. A possible reason for this finding concerns

cohort age: few interface design evaluations have included elderly people, with only younger people being tested in one previous study [3]. In a pilot test, elderly people spent more time on such tasks than did younger people, due to an age effect [14]. As only elderly subjects were tested, it is unclear whether the aging effect could have masked the gender effect, because younger people were not tested. Based on the gender-effect finding, the first hypothesis – gender will significantly affect the comparative ratings of digital dictionary interface usability given by elderly people – was not supported, likely due to being masked the effect of age [15]. A further experiment including younger people is recommended.

The effect of mode (vertical or horizontal) was also not significant. Although people may show a preference for either a vertical or a horizontal interface layout, a mode effect was not anticipated to ensure the quality of the interaction. The existence of a mode effect for any given interface would suggest that further modifications or improvements are required.

A significant effect of topology was found, consistent with most previous studies [11]. However, the present study not only emulated previous studies with respect to finding a topology effect, but also modified the way in which this effect was tested for by using a mixed type topology that incorporated the strengths of both hierarchal and network topological interface designs. The mixed type topology was superior to the linear interface design. Thus, Hypothesis 2 – topology will significantly affect the comparative ratings of digital dictionary interface usability given by elderly subjects – was supported by the present findings.

A significant difference in performance effectiveness was found according to task type. Pairwise comparisons revealed the following task order with respect to search time: easy < medium < difficult task. This result is consistent with a previous study, in which more difficult tasks took more time [13, 16]. As the easy task was simple and single-layered, it was completed by all users in less time; task time increased accordingly for the medium (double-layered) and difficult (triple-layered) tasks.

4.2. Interaction between task complexity and topological structure

The experimental data indicated that the main effects of topological structure and task effect, and their

interaction, were significant with respect to task performance (Fig. 7). A post hoc comparison revealed that the mixed type topology was superior to the linear topological interface design for all three task types, and a positive correlation was found between topology and task variables. Thus, more difficult tasks were characterized by larger differences in the operation performance between the mixed and linear topological structures. Therefore, Hypothesis 3 – gender, task type, and topological structure will have an interaction effect on interface operation performance – was supported by the present results.

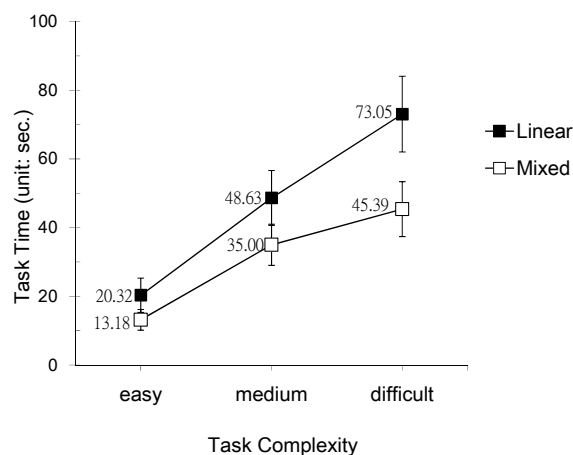


Figure 7. Mean task time scores for the two topological structures for each task type

Although most studies have proposed an interaction between topological structure and task variables [17, 18], it is important to note the fundamentally different approach of the present versus previous studies. Based on the findings of previous studies, the present study extended the design to encompass an interactive effect, by proposing a mixed design that integrates hierarchical and network topologies. A previous study [11] argued that a linear design would result in better user performance in a simple task, whereas other design would be superior for more difficult tasks; this was not found in the present study. A possible reason is that appropriate software interfaces encourage elderly people to use digital apps [19], which is why the mixed type of interface design showed better performance than the linear design; the former featured reduced clutter and terminology on the screen, and clear and simple navigation paths were provided for the user. These design considerations are important for all users, but particularly for the elderly [20].

This study extended the assessment of interface design by including a mixed design that integrated the

advantages of both hierarchical and network topological structures, and demonstrated its utility for elderly users. The pattern of interactive effect was changed, thus, a positive ratio effect was detected between topology and task factors. However, other potential variables that may also affect the effectiveness and usability of an interface, such as age [15] or wayfinding strategy [21], were not changed. Future studies should use an advanced experimental design to explore the effects of other, related factors, to improve interface design for elderly people and other user groups.

It is important to note that this study can benefit the application of the Chinese/ English digital translator or related app interface design, but the comparison of different age group or language such as Japanese/ English which can be planned in the future to further explore the correlation between different languages. For example, it was proved that gender and display mode did not affect the task performance regardless interface design or task complexity, which may be changed due to different age groups or languages. Further studies are recommended to investigate whether these findings can be generalized to different age groups and languages.

5. Conclusions

In this study, different topological designs were examined, and the main effects of gender and mode on task performance were revealed, not significant, and neither were the gender interaction effects with all other factors. It was also found that the main effects of topological structure and task effect, and their interaction, were significant with respect to task performance. An example of a good interface design (mixed type) that can be adapted to suit particular task conditions and allows high-quality operation by elderly users was proposed by this study. The mixed topological design improved task performance relative to a linear topological interface design, regardless of task complexity. The findings of this study may contribute to improvements in digital dictionary app interface design and research for elderly people, and may be facilitative in terms of assessing changes in interface design for different user groups or languages over time.

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